Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Lesson 1: Model Selection Techniques

- Introduction to Model Selection
  Reading • 5 min
- Bayesian Model Selection
  Reading • 30 min
- Bayesian Model Selection Readings
  Reading • 1h

Lesson 2: Cross Validation

- Introduction to Cross-Validation
  Reading • 25 min
- K-Fold Cross-Validation
  Reading • 2 min
- Leave-One-Out Cross-Validation (LOOCV)
  Reading • 10 min

Lesson 3: Introduction to Optimization Techniques

- Introduction to Optimization Techniques
  Reading • 13 min
- Optimization Using Gradient Descent
  Reading • 40 min
- Gradient Descent with Momentum
  Reading • 45 min
- Stochastic Gradient Descent (SGD)
  Reading • 40 min
- Constrained Optimization
  Reading • 16 min
- Lagrange Multipliers
  Reading • 8 min
- Convex Optimization
  Reading • 8 min
- Linear Programming
  Reading • 15 min
- Quadratic Programming
  Reading • 10 min
- Check Your Knowledge: Optimization Techniques
  Graded Assignment • 15 min

Bayesian Model Selection

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Bayesian technique is a framework for comparing different models to determine which best explains observed data, using the principles of Bayesian probability where the goal is to update our beliefs about the plausibility of different models as we observe data. It evaluates models not only based on how well they fit the data but also by considering model complexity.

Key Components of Bayesian Model Selection

1. **Likelihood** $P(\mathcal{D}|\theta, M)$:
   - Represents how likely the observed data $\mathcal{D}$ is under a given model M and its parameter/s θ, if applicable.
   - A model with higher likelihood fits the data better.
2. **Prior Probability of the Model** $P(M)$:
   - Encodes prior beliefs about the plausibility of a model before observing the data.
   - For example, if we believe simpler models are more plausible, we assign them higher prior probabilities.
3. **Prior Probability of the parameters** $P(\theta|M)$
   - Prior probability of the parameters under model M.
4. **Model Evidence (Marginal Likelihood)** $P(\mathcal{D}|M)$:
   - The probability of observing the data under a model, integrating over all possible parameter values: $P(\mathcal{D}|M) = \int P(\mathcal{D}|\theta, M)P(\theta|M)d\theta$.
   - It captures both the fit of the model and its complexity, penalizing overly complex models.
5. **Posterior Probability** $P(M|\mathcal{D})$:
   - The probability of a model given the data, calculated using Bayes' theorem: $P(M|\mathcal{D}) = \frac{P(\mathcal{D}|M)P(M)}{\sum_i P(\mathcal{D}|M_i)P(M_i)}$.
6. **Bayes Factor**:
   - A measure to compare two models $M_1 and M_2$, calculated as the ratio of their evidences: $BF_{12} = \frac{P(\mathcal{D}|M_1)}{P(\mathcal{D}|M_2)}$.
   - If:
     - $BF_{12} > 1$: Evidence favors M_1.
     - $BF_{12} < 1$: Evidence favors M_2.
   - Posterior Odds are the ratio of the posterior probabilities of the two models: $\frac{P(M_1|\mathcal{D})}{P(M_2|\mathcal{D})}$
   - The posterior odds can be broken down into two parts: $\frac{P(M_1|\mathcal{D})}{P(M_2|\mathcal{D})} = \frac{P(M_1)}{P(M_2)} \cdot \frac{P(\mathcal{D}|M_1)}{P(\mathcal{D}|M_2)}$

Steps in Bayesian Model Selection

1. **Define Models**:
   - Specify the models to be compared, including their likelihood functions and parameter priors.
2. **Compute Marginal Likelihood (Evidence)**:
   - For each model, calculate $P(\mathcal{D}|M)$ by integrating the likelihood over the prior distribution of parameters.
3. **Compare Models**:
   - Use the Bayes factor to quantify the relative support for each model.
4. **Interpret Results**:
   - Select the model with the highest posterior probability or evidence, considering the strength of the Bayes factor.

Advantages of Bayesian Model Selection

1. **Handles Model Complexity**:
   - Automatically penalizes overly complex models by averaging over parameter spaces, reducing the risk of overfitting.
2. **Incorporates Prior Knowledge**:
   - Allows the inclusion of prior beliefs about parameters or models.
3. **Probabilistic Interpretation**:
   - Provides probabilities for models, making the results interpretable and meaningful.

Example: Coin Toss

Suppose you observe 10 coin tosses with 7 heads and 3 tails. You are comparing:

1. M_1: The coin is fair $P(\text{Heads}) = 0.5$ using binomial distribution with parameter $p = 0.5$.
2. M_2: The coin is biased $P(\text{Heads}) = p$, where p is unknown.

This is your dataset $\mathcal{D} = \{H = 7, T = 3\}$.

Marginal Likelihood for M_1:

$P(\mathcal{D}|M_1) = \binom{10}{7}(0.5)^{10} = 0.117$

Marginal Likelihood for M_2:

$P(\mathcal{D} \mid P, M_2) = p^7(1-p)^3$.

Since p is unknown, we integrate over all possible values of p, weighted by a prior distribution. Assume a **uniform prior** for $P(p) = 1 for p \in [0, 1]$

$P(\mathcal{D}|M_2) = \int_0^1 p^7(1-p)^3 dp = 0.00076$

Using the Bayes factor:

$BF_{12} = \frac{P(\mathcal{D}|M_1)}{P(\mathcal{D}|M_2)}$

$BF_{12} = \frac{0.117}{0.00076} \approx 154.0$

$BF_{12} > 1$: Evidence favors M_1 (the fair coin model).

A Bayes factor of 154.0 means the data is **154 times more likely** under the fair coin model than under the biased coin model.

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